4 self in Python



KEY IDEAS

- `self` is the **instance the method was called on**, passed in automatically as the first argument. A naming convention, not a keyword; always call it `self`.
- `obj.method(x)` is sugar for `Class.method(obj, x)`. `obj.method` is a **bound method** with `self` already filled in.
- Use `self.x` for instance state. `return self` enables method chaining.

PITFALL Writing `def hello():` (no `self`) gives `TypeError: ... takes 0 positional arguments but 1 was given`. Inside a method a bare `name` is a local variable; instance state needs `self.name`.

EXAMPLE

```
class Greeter:
    def __init__(self, name):
        self.name = name

    def hello(self):
        return f"Hi, {self.name}"

    def rename(self, new):
        self.name = new
        return self                 # allows chaining

g = Greeter("Ann")
print(g.hello())                    # → Hi, Ann
print(Greeter.hello(g))             # → Hi, Ann  (same call)
print(g.hello.__self__ is g)        # → True
print(g.rename("Bob").rename("Cy").hello())
# → Hi, Cy
```